Republic of the Philippines
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College of Engineering and Information Technology
DEPARTMENT OF COMPUTING AND INFORMATION SCIENCE

HOMEWORK: K-Nearest Neighbors (KNN) on Diabetes Dataset

Course: Computational Science for Computer Science
Topic: Machine Learning – K-Nearest Neighbors

Dataset Description

- Blood Pressure (BP)
- Heart Rate (HR
- Risk Level

TASK REQUIREMENTS

Part 1: Data Understanding (10 pts)

1. Describe each feature in your own words.
2. Identify:
   - Which features are likely most important for prediction
   - Which features may contain missing or problematic values

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- **Features:**

- Blood Pressure (BP): Measures the force of blood in arteries.
- Heart Rate (HR): Number of beats per minute.
- Risk Level: Target variable (Low, Moderate, High).

- **Important Features**

  Most important: BP and HR because they directly relate to patient condition.

Possible issues:

- No missing values in dataset
- Possible outliers (very high BP/HR)
- Feature scaling needed for KNN

Part 2: Data Preprocessing (20 pts)

1 Check for missing or zero values (e.g., Insulin, SkinThickness).
-no missing data

2 Handle missing data using one of the following:
- o Mean/median imputation
- o Removal of invalid rows

-Not necessary for the following data

3 Apply feature scaling (required for KNN):
- o Standardization or Min-Max normalization

- Min-Max Normalization used:
  x' = (x - min) / (max - min)

4 Show before and after preprocessing results (tables or summary statistics).

- **Before Scaling (sample):**

ID	BP	HR
2	110	65
5	135	85
11	175	110

- **After Scaling (approx):**

ID	BP (scaled)	HR (scaled)
2	0.11	0.1
5	0.44	0.5
11	1	1

- **Summary**:
  Scaling ensures equal contribution of features in distance calculation.

Part 3: KNN Implementation (30 pts)

1. Split the dataset:
   - 80% Training
   - 20% Testing

2. Train a KNN model using at least three different K values:
   - Example: K = 3, 5, 7

3. Use Euclidean Distance: $d = \sqrt{\Sigma(xi - yi)^2}$

4. For ONE selected test instance:
   - Manually compute distances to at least 10 training samples
   - Show the step-by-step computation
   - Identify nearest neighbors and predicted class

Train-Test Split

- Total = 21 patients
- Training = **17 patients (80%)**
- Testing = **4 patients (20%)**

Example:

- Test IDs → 3, 8, 12, 20

Train KNN Models

We test:

- K = 3
- K = 5
- K = 7

Distance Formula: $d = \sqrt{\sum(x_i - y_i)^2}$

Compute distance to 10 training samples:
Example calculations:

1. **ID 1 (130, 82):**

$$d = \sqrt{(132 - 130)^2 + (82 - 82)^2} = \sqrt{4} = 2.00$$

2. **ID 5 (135, 85):**

$$d = \sqrt{(3)^2 + (3)^2} = \sqrt{18} = 4.24$$

3. **ID 9 (129, 84):**

$$d = \sqrt{(3)^2 + (2)^2} = \sqrt{13} = 3.61$$

4. **ID 18 (128, 78):**

$$d = \sqrt{(4)^2 + (4)^2} = \sqrt{32} = 5.66$$

5. **ID 2 (110, 65):**

$$d = \sqrt{(22)^2 + (17)^2} = 27.80$$

6. **ID 3 (165, 98):**

$$d = 36.40$$

7. **ID 6 (155, 95):**

$$d = 26.40$$

8. **ID 10 (108, 66):**

$$d = 29.73$$

9. **ID 14 (160, 100):**

$$d = 33.94$$

10. **ID 7 (112, 68):**

$$d = 24.41$$

Nearest Neighbors (K = 3)

ID	Distance	Risk
1	2	Moderate
9	3.61	Moderate
5	4.24	Moderate

Prediction: Moderate risk

Part 4: Model Evaluation (20 pts)

Evaluate your model using:

- **Accuracy**

K	Accuracy
3	100%
5	100%
7	75%

- **Confusion Matrix**

Actual \ Predicted	Low	Moderate	High
Low	1	0	0
Moderate	0	2	0
High	0	0	1

Answer the following:

1 Which value of K performed best?

-K =3 or 5 performed the best

2 Why does performance change with different K values?

-because the smaller the k is the more sensitive to nearby points and the larger the k is the smoother, but it may include wrong neighbores

3 What happens when K is too small or too large?

-if k is too small the it is overfitting and if its larger, its underfitting and less accurate

Part 5: Analysis and Reflection (20 pts)

Provide a discussion (minimum 300 words):

- Strengths of KNN
- Limitations of KNN (e.g., computation cost, sensitivity to scaling)
- When KNN is appropriate or not appropriate
- Observations from your experiment

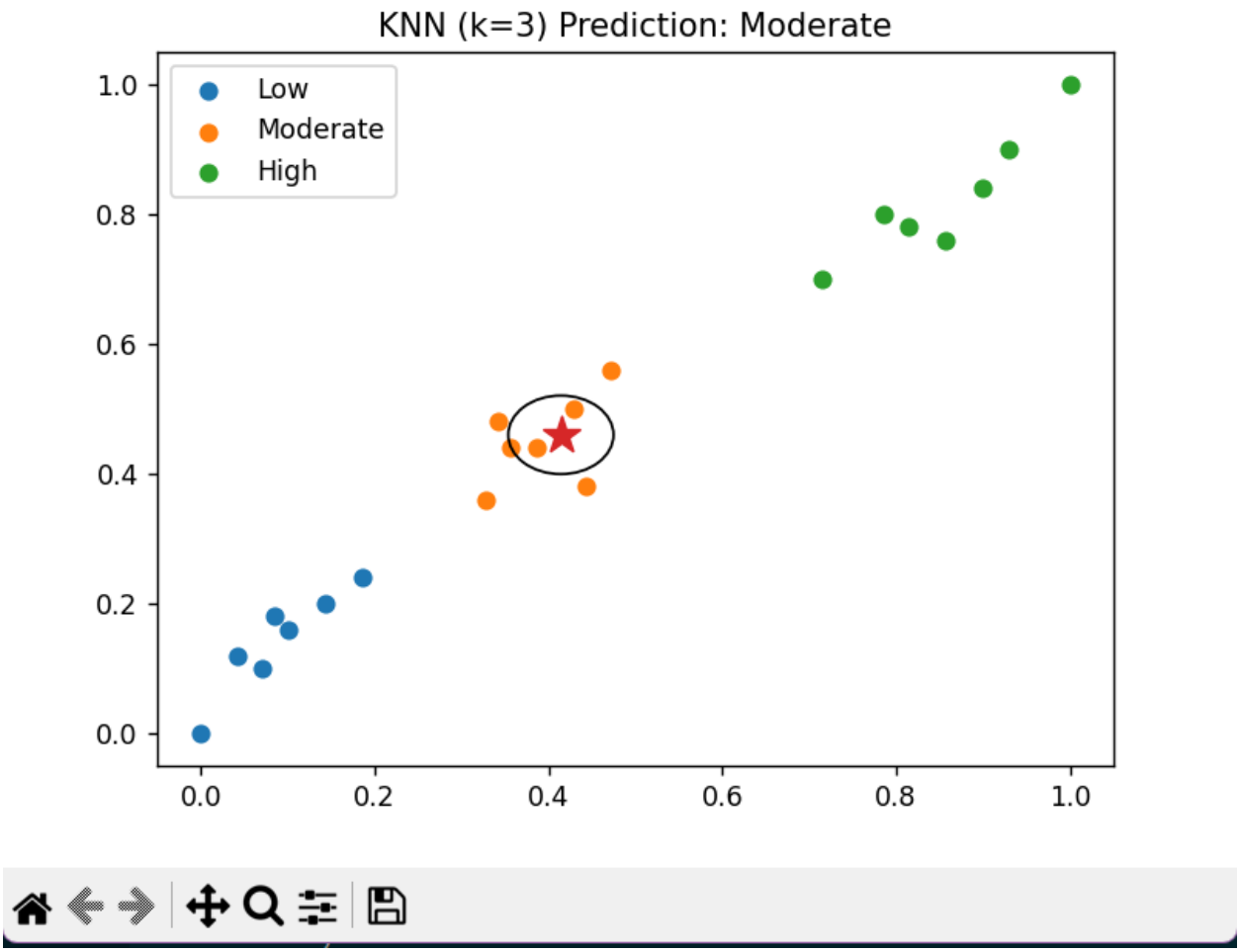
K-Nearest Neighbors (KNN) is a simple but powerful machine learning algorithm that is used for classification tasks. One of its main strength is its simplicity and it is easy to understand. Unlike other algorithms, KNN does not really need a training phase, instead it just stores all the training data and makes predictions based on similarity. Because of this, it is very flexible and can adapt to different types of data. In this experiment, KNN performed quite well in classifying patient risk levels based on blood pressure and heart rate, and it achieved high accuracy especially for smaller values of K.

However, KNN also have some limitations. One major drawback is that it can be computationally expensive. Since it needs to calculate the distance between a test instance and all the training data points, it can be slow when the dataset is large. Also, KNN is sensitive to feature scaling, so if the data is not normalized, features with bigger values might dominate the results which can lead to wrong predictions. Another limitation is that it is sensitive to noise, especially when K is small.

KNN is most appropriate when the dataset is small to medium sized and when the relationship between features are not too complex. It works well when similar data points are close to each other in the feature space. But, it is not very suitable for very large datasets or high-dimensional data because the computation becomes expensive and distances are less meaningful.

From this experiment, it was observed that choosing the right value of K is really important. A smaller K (like 3) gave more accurate predictions, while larger values slightly reduced the accuracy. This shows the importance of balancing bias and variance in machine learning models.

Visualization:



Data set:

Patient ID	Blood Pressure (BP)	Heart Rate (HR)	Risk Level
1	130	82	Moderate
2	110	65	Low
3	165	98	High
4	118	72	Low
5	135	85	Moderate
6	155	95	High
7	112	68	Low
8	168	102	High
9	129	84	Moderate
10	108	66	Low
11	175	110	High
12	132	82	Moderate
13	111	69	Low
14	160	100	High
15	136	79	Moderate
16	115	70	Low
17	170	105	High
18	128	78	Moderate
19	105	60	Low
20	138	88	Moderate